MODULE 34 · FOUNDATIONS AND EXAM SKILLS - ALL DOMAINS

Confusion pairs

The services and settings most often mixed up, each reduced to the single question that separates them.

DOMAIN Exam skills WEIGHT - MODULE 34 of 39

Every confusion pair has **one question** that separates them. Learn the question, not the products.

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Why these pairs keep appearing

PATTERN	EXAMPLE
Same family, different job	Image Analysis and a multimodal model
Adjacent capability	Speech and Translator
Indexing time versus query time	Embedding skill and vectorizer
Direction of control	Handoff and agent-as-tools
Retired or legacy versus current	Prompt flow and Agent Framework

ON THE EXAM

Recognising the pattern is often faster than recalling the products. Ask what kind of difference is being tested.

Document Intelligence or Content Understanding?

DOCUMENT INTELLIGENCE

Train on what you have

- Stable forms, consistent layouts
- Custom models need labelled samples
- Deterministic, structured extraction

CONTENT UNDERSTANDING

Describe what you want

- Varied templates and languages
- Zero-shot: fields in plain language
- Markdown and grounded fields for RAG

ON THE EXAM

Ask: do I have labelled samples and stable layouts, or many templates and a description of what I need?

Image Analysis or a multimodal model?

IMAGE ANALYSIS

Coordinates and labels

- Objects and regions with bounding boxes
- Tags and short captions
- Deprecated, retiring September 2028

MULTIMODAL MODEL

Description and reasoning

- Captions, alt text, visual question answering
- Reasoning about what an image shows
- No bounding boxes

ON THE EXAM

Ask: do I need to know where something is, or what it means?

Custom Vision or Image Analysis?

CUSTOM VISION

Your own classes

- Trained on your labelled images
- Detects your specific products or parts
- Microsoft's pointer for custom object detection

IMAGE ANALYSIS

Everyone's classes

- Prebuilt categories and common objects
- No training required
- Broad, generic recognition

ON THE EXAM

Ask: are the classes mine, or the world's? Both retire on 25 September 2028.

Speech or Translator?

SPEECH

Anything spoken

- Transcription, synthesis, speech translation
- Live Interpreter for language switching
- Audio in, audio or text out

TRANSLATOR

Anything written

- Text and document translation
- Tone, gender and adaptive terminology
- Text in, text or file out

ON THE EXAM

Ask: is the input audio? Then Speech first. Translator never hears anything.

File search, AI Search tool, or Foundry IQ?

FILE SEARCH

You hand over files

- Upload and the service chunks and embeds
- Up to 10,000 files per vector store
- Fastest to stand up

AI SEARCH TOOL AND FOUNDRY IQ

An index or a knowledge layer

- AI Search tool: query an index you run
- Foundry IQ: many sources, permission-aware
- Agentic retrieval and citations

ON THE EXAM

Ask: whose index is it? Nobody's yet, one you already run, or many sources with permissions.

Prompt agent or hosted agent?

PROMPT AGENT

Configuration only

- Model, instructions, built-in tools
- Foundry runs it; nothing to host
- Fastest path to a working agent

HOSTED AGENT

Your code, hosted

- Your framework: Agent Framework, LangGraph
- Custom protocols and compute control
- State between turns in your own code

ON THE EXAM

Ask: is there custom orchestration code? No means prompt agent.

Handoff or agent-as-tools?

HANDOFF

Control moves

- The receiving agent takes over
- No central authority
- Triage to a specialist who finishes

AGENT-AS-TOOLS

Control returns

- Results come back to the primary agent
- The primary keeps responsibility
- One assistant combining experts

ON THE EXAM

Ask: who owns the conversation at the end? The A2A tool behaves like agent-as-tools.

MCP or A2A?

MCP

Agent to tools

- Exposes tools and data
- The tool does as it's told
- Your agent does the reasoning

A2A

Agent to agent

- Connects to another agent
- That agent reasons for itself
- Discovered through an agent card

ON THE EXAM

Ask: does the other side think? A search index doesn't. A partner's booking agent does.

Conversation or memory?

CONVERSATION

The transcript

- Every message and tool call in one dialogue
- Continue the same case later
- Grows with each turn

MEMORY

The notes

- Facts and preferences about a person
- Carried into new conversations
- Scoped per user

ON THE EXAM

Ask: continuing this case, or recognising this person in a new one?

Groundedness, relevance or similarity?

WHAT THEY NEED

Different inputs

- Groundedness needs the retrieved context
- Similarity needs a reference answer
- Relevance needs only question and response

WHAT THEY CATCH

Different failures

- Groundedness: invented content
- Relevance: off-topic answers
- Similarity: differs from a known answer

ON THE EXAM

Ask what you have: context, references, or neither. That eliminates two of the three immediately.

Embedding skill or vectorizer?

EMBEDDING SKILL

Indexing time

- Vectorises documents as they're indexed
- Part of the skillset
- Runs once per document

VECTORIZER

Query time

- Embeds the incoming question
- Defined on the index, used by a vector profile
- Runs on every query

ON THE EXAM

Ask which side isn't embedded yet. Missing vectorizer means the app must embed queries itself.

HNSW or exhaustive KNN?

HNSW

Approximate

- Scales to large indexes
- Uses vector index quota
- The production default

EXHAUSTIVE KNN

Exact

- Brute-force scan of every vector
- No vector index quota used
- Small sets, and testing recall

ON THE EXAM

Ask: does this need to be exact, or fast at scale? You can override to exhaustive at query time.

SFT, DPO or RFT?

WHAT YOU HAVE

Decides the method

- Labelled input and output pairs: SFT
- Preferred versus rejected pairs: DPO
- A grader that scores results: RFT

WHAT IT FIXES

Different gaps

- SFT: the task itself
- DPO: which of two good answers is better
- RFT: complex behaviour worth rewarding

ON THE EXAM

Ask what your data looks like. The data you already have usually names the method.

One question per pair

PAIR	ASK
Document Intelligence / Content Understanding	Labelled samples, or a description?
Image Analysis / multimodal model	Where it is, or what it means?
Custom Vision / Image Analysis	My classes, or the world's?
Speech / Translator	Is the input audio?
Prompt agent / hosted agent	Is there custom code?
Handoff / agent-as-tools	Who owns the conversation at the end?
MCP / A2A	Does the other side think?
Embedding skill / vectorizer	Which side isn't embedded yet?

ON THE EXAM

Eight questions cover most of the pairs the exam builds distractors from.

> A manufacturer photographs assembly lines and must locate twelve of its own machine parts in each photo, with boxes drawn around them for operators.
>
> The parts are specific to this company, and thousands of labelled photos already exist.

What should the team use?

A Image Analysis object detection, which returns bounding boxes already.

B Custom Vision, trained on the company's labelled photos of its parts.

C A multimodal model, asked to describe where each part appears.

D Content Understanding, with a field schema naming the twelve parts.

◆ PAUSE AND DECIDE

What should the team use?

A Image Analysis objects.
Detects common objects, not this company's specific parts.

B Custom Vision.
Custom classes from labelled photos, with boxes: Microsoft's pointer for custom object detection.

C Multimodal model.
Describes and reasons, but returns no bounding boxes.

D Content Understanding.
Extracts fields from content; it isn't an object detector.

THE SIGNAL

Two questions, answered in order: my classes, not the world's; and where, not what.

> A team has an existing LangGraph application with its own orchestration and a custom message protocol for a partner system. They want it running in Foundry, with identity, scaling and tracing.
>
> They don't want to rewrite the orchestration.

What should they deploy?

A. A prompt agent, with the orchestration expressed in its instructions.

B. A workflow agent, rebuilt in the portal's visual designer.

C. A knowledge base, with the LangGraph code called as a tool.

D. A hosted agent, packaging their existing code in a container.

◆ PAUSE AND DECIDE

What should they deploy?

A Prompt agent.

Configuration only. It can't run their orchestration code or custom protocol.

B Workflow agent.

A rewrite, into a designer that retires in December 2026.

C Knowledge base.

Grounding data, not a place to run application code.

D Hosted agent.

Their framework and protocol, with Foundry identity, scaling and tracing.

THE SIGNAL

'Is there custom code?' answers this in one step. Custom protocols reinforce it.

> A retail agent should greet returning customers by recalling their size and dietary preferences, even when they open a brand-new chat about a different product months later.
>
> Each chat about one order already continues correctly within itself.

What does the team need to add?

A. Memory, scoped to each signed-in customer, storing profile facts.

B. A longer conversation, reused across every chat that customer starts.

C. A larger context window, so older turns still fit in each request.

D. File search over exported chat transcripts for every customer.

◆ PAUSE AND DECIDE

What does the team need to add?

A Per-user memory.

Facts about a person, carried into brand-new conversations.

B One long conversation.

Mixes unrelated cases into one transcript that grows forever.

C Larger context window.

Fits more of one conversation; carries nothing between them.

D File search on transcripts.

Turns personal history into a searchable corpus: costly and a privacy risk.

THE SIGNAL

'Recognising this person in a new case' is memory. 'Continuing this case' is the conversation.

> A media library indexes 40 million chunk vectors, and its search must answer within a few hundred milliseconds.
>
> Occasional approximate matches are acceptable; slow responses are not.

Which vector algorithm fits?

A Exhaustive KNN, because exact neighbours give the best relevance.

B Exhaustive KNN, because it doesn't consume vector index quota.

C HNSW, with an exhaustive override available for testing recall.

D Neither: use keyword search with a scoring profile instead.

◆ PAUSE AND DECIDE

Which vector algorithm fits?

A Exhaustive for relevance.

Exact, but a brute-force scan of 40 million vectors won't meet the latency budget.

B Exhaustive for quota.

A real advantage, chosen for the wrong reason: latency is the constraint.

C HNSW, with exhaustive for testing.

Approximate search at scale, with exact comparison available when testing.

D Keyword only.

Abandons meaning-based matching the library depends on.

THE SIGNAL

Exact or fast at scale? Two of the four options pick the same algorithm for different reasons: read why, not just what.

Confusion pairs in five lines

- **Learn the separating question**, not the feature lists.
- **Where or what; mine or the world's; audio or text.**
- **Custom code or not; control returns or moves; tools or agents.**
- **Indexing time or query time; exact or fast; context, references or neither.**
- **Read why an option is offered**, not just which product it names.

> ON THE EXAM
>
> Next, module 35: reading Foundry Python code, since about a quarter of bank questions carry code.